1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist highlighted that docetaxel-treated prostate cancers show divergent transcriptome remodeling and resistance phenotypes—why do some tumors become docetaxel-resistant despite exposure? 2) Tier 1 paths used: I built on the MGMT mRNA path (path 1) because it offers a provocative, measurable resistance readout, and the full-length AR mRNA/AR path (path 2) because AR transcript integrity is a central, mechanistic determinant of prostate tumor state and therapy sensitivity. These two anchors directly suggest a cell-intrinsic repair/signaling axis that could enable survival under taxane pressure. 3) Tier 2 support: Evidence that docetaxel alters transcriptomes (transfrags from DU145 RNA-seq) and that epigenetic reprogramming (DZNep → KRT7 P1) can unmask alternative transcripts supports a model where microenvironmental or epigenetic states change transcript architecture and repair programs, reinforcing AR/MGMT-driven resistance. 4) Why therapy_response endpoint: The paths explicitly concern treatment (docetaxel) and resistance; a therapy_response proxy is the most direct, actionable clinical endpoint for testing mechanistic links between tumor state, transcripts, and drug outcome. 5) Deliberate exclusions: I excluded the software/data-pipeline node (Oases) and any broad, multi-gene predictive lists to keep the hypothesis sharply mechanistic (max 2 anchor genes). SOX2, transfrags, KRT7 P1, DU145, and DZNep were placed in secondary_modifiers for follow-up rather than in the core claim to avoid diluting the testable question?

Working Hypothesis

Do tumors with a high stromal-to-tumor ratio (TSR) resist docetaxel because a stromal-rich microenvironment preserves full-length AR mRNA and upregulates MGMT mRNA–driven stress-repair programs that enable survival under taxane pressure? If so, TSR-high tumors with elevated AR and MGMT mRNA expression will show poorer therapy response to docetaxel, identifying patients who may benefit from AR-directed or repair-targeted combination strategies.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	AR, MGMT mRNA
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

AR: AR mRNA/AR path (path 2) because AR transcript integrity is a central, mechanistic determinant of prostate tumor state and therapy sensitivity.

MGMT mRNA: MGMT mRNA path (path 1) because it offers a provocative, measurable resistance readout, and the full-length AR mRNA/AR path (path 2) because AR transcript integrity is a central, mechanistic determinant of prostate tumor state and therapy sensitivity.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	SOX2	Supporting modifier from knowledge graph
Secondary	transfrag	Supporting modifier from knowledge graph
Secondary	KRT7 P1 transcript	Supporting modifier from knowledge graph
Secondary	DZNep	Supporting modifier from knowledge graph
Secondary	DU145 cells	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: quantify TSR in tumor samples and categorize TSR-high vs TSR-low;

Prediction 2: measure AR full-length mRNA and MGMT mRNA levels;

Prediction 3: in vitro and ex vivo models (including DU145-derived systems and stromal co-cultures) to assess docetaxel sensitivity with and without modulation of AR and MGMT;

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SOX2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- transfrag: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.